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Gut microbiota-derived gamma-aminobutyric acid from metformin treatment reduces hepatic ischemia/reperfusion injury through inhibiting ferroptosis

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Abstract

Hepatic ischemia/reperfusion injury (HIRI) is a common and inevitable factor leading to poor prognosis in various liver diseases, making the outcomes of current treatments in clinic unsatisfactory. Metformin has been demonstrated to be beneficial to alleviate HIRI in recent studies, however, the underpinning mechanism remains unclear. In this study, we found metformin mitigates HIRI-induced ferroptosis through reshaped gut microbiota in mice, which was confirmed by the results of fecal microbiota transplantation (FMT) treatment but showed the elimination of the beneficial effects when gut bacteria were depleted using antibiotics. Detailedly, through 16S rRNA and metagenomic sequencing, we identified that the metformin-reshaped microbiota was characterized by the increase of gamma-aminobutyric acid (GABA) producing bacteria. This increase was further confirmed by the elevation of GABA synthesis key enzymes, glutamic acid decarboxylase (GAD) and putrescine aminotransferase (PAT), in gut microbes of metformin-treated mice and healthy volunteers. Furthermore, the benefit of GABA against HIRI-induced ferroptosis was demonstrated in GABA-treated mice. Collectively, our data indicate that metformin can mitigate HIRI-induced ferroptosis by reshaped gut microbiota, with GABA identified as a key metabolite.

eLife assessment

This study presents a **valuable** finding on the impact of metformin-induced shifts in gut microbial community structure and metabolite levels for drug efficacy in a mouse model of liver injury. The current evidence supporting the claims of the authors is **incomplete**, although inclusion of additional controls and a revision to clarify the reviewer's methodological concerns could strengthen the study. With revision, this paper could be of broad interest to researchers across multiple disciplines, including the microbiome, liver disease, and pharmacology.

Introduction

Hepatic ischemia/reperfusion injury (HIRI) remains a major cause of liver damage mainly following liver transplantation, complicated hepatectomy, haemorrhagic shock and severe liver trauma.^{1–4} Traditional therapeutics have been developed and repurposed to treat HIRI, including ischemic preconditioning, machine perfusion technologies, gene interventions, and stem cell therapy.^{5–8} Unfortunately, current treatments are not satisfactory due to the complicated pathogenesis of HIRI. Recently, a novel theory relating to the gut-liver axis has been well accepted, demonstrating that gut microbiota is vital for liver diseases, however, there is no strategy targeted this theory to alleviate HIRI in clinical practice.

The liver links with the intestine by portal circulation which allows the transfer of gut-derived products, not only nutrients, but also microbial metabolites and components.⁹ As an important part of the gut-liver communication, the intestinal mucosal and vascular barriers limit the systemic dissemination of microbes and toxins while allowing nutrients to access the circulation and reaching the liver.¹⁰ Additionally, it is generally believed that liver injury can provoke intestinal mucosal damage and inflammation, further resulting in gut dysbiosis, which reversely induces a secondary attack on the liver.¹¹ Microbial agents, such as synbiotics and probiotics, have demonstrated to attenuate different liver diseases in animal studies,^{12, 13} indicating that modulating gut microbes potentially prevents HIRI in clinical practice.

Recently, the hypoglycaemic drug, metformin, has attracted intensive attention in its pleiotropic biological actions in various diseases, such as obesity, cancer, neuroinflammation, especially liver diseases.^{14–17} As evidences shown, the gut with high concentration of metformin become an important action site to reshape microbiota.¹⁸ According to *Sun et al.*,¹⁹ metformin was proven to decrease the abundance of the *B. fragilis* species and to increase the levels of bile acid and glycoconjugated bile acid in the gut, which inhibited intestinal farnesoid X receptor against obesity-related metabolic dysfunction in mice. Moreover, it is also demonstrated that metformin could attenuate sepsis-related liver injury by increasing the proportion of *Bifidobacterium*, *Muribaculaceae*, *Parabacteroides distasonis*, and *Alloprevitella* in rats.²⁰ As for the HIRI, metformin has also shown the protective effect,²¹ but the detailed role of gut microbiota remains unclear.

Although metformin showed protective effects on HIRI, the role and distinctions of reshaping microbes have yet to be determined. In the present study, we sought to elucidate that gamma-aminobutyric acid (GABA)-producing bacteria played a critical role for the protective effects of metformin against HIRI by using 16S rRNA and metagenomic sequencing to find the different features of gut microbiota. The finding was reconfirmed by the results of mice that were treated with metformin-reshaped fecal microbiota or oral antibiotics administration. Furthermore, GABA was found to show a protective effect on HIRI by intraperitoneally injection into mice in advance,

while an increase of microbial GABA was observed in both mice and human experiments after metformin treatment. Evidenced by the improvement of HIRI, we hypothesize that metformin-reshaped gut microbiota elicits greater therapeutic responses for liver diseases and serve as a promising targeted therapeutic.

Results

Metformin significantly mitigates HIRI and reshapes gut microbiota in mice

In order to evaluate the protective effect of metformin on HIRI, survival experiment was employed in mice. The survival rate of HIRI mice increased from 41.67% to 90.00% after metformin intervention (**Fig. 1a**), accompanied by hepatic morphological and histopathological improvement including the recovery of liver pathological structure, the decrease in red blood cell aggregation and consumption of liver glycogen (**Fig. 1b**), as well as the decrease in serum activity of ALT and AST (**Fig. 1c**). Qualification of MDA and GSH concentrations in the mice liver demonstrated that the presence of metformin significantly attenuated HIRI-induced oxidation (**Fig. 1d**).

In order to clarify the critical role of gut microbiota in the pleiotropic actions of metformin,^{22–24} fecal samples were collected from the mice to perform 16S rRNA sequencing. It was found that the gut microbiota was substantially modulated in IR + metformin (Met) group, showing decreased abundance (alpha diversity) and reshaped distribution (**Fig. 1e**). Next, as the microbiota-gut-liver axis theory indicates,²⁵ HIRI-induced dysfunction of the gut barrier may aggravate liver damage by disrupting the gut microbiota. H&E staining of the colon and ileum, mainly involved in gut barrier dysfunction, showed that the density and integrity of intestinal mucosa were strikingly reduced by HIRI, which was restored by metformin treatment (**Fig. 1f**). The immunofluorescence analysis of colon tissues showed that the decreased tight junction protein ZO-1 and Occludin in HIRI-treated mice were reversed by metformin (**Fig. 1g** and **1h**). Similarly, western blot analysis of Occludin in colon (**Fig. 1g**) and the mRNA levels of *Occludin*, *ZO-1*, *Claudin-1*, *JAM1* and *JAM4* in ileum of HIRI mice were increased with metformin administration (**Fig. 1i**). The inflammatory cytokine *IL-6*, *IL-1 β* and *IL-18* were significantly upregulated in the ileum and colon received HIRI and were reduced by metformin treatment (**Fig. 1j**). In addition, the serum FD-4 level in IR group was cut down by metformin (**Fig. 1k**). These results suggested that metformin plays a critical role in restoring gut barrier function and alleviating HIRI.

Metformin-reshaped fecal microbiota attenuates HIRI

In order to further confirm the role of metformin in reshaping the gut microbiota, microbes were transplanted into mice before inducing reperfusion (IR). As expected, the survival analysis of mice in the IR + fecal microbiota transplantation (FMT) group showed a similarly protective efficacy as the IR + Met group, as indicated by the significantly improved survival rate of IR mice (**Fig. 2a**). This finding was validated by ALT and AST levels as well as H&E, PAS and DHE staining (**Fig. 2b** and **2e**). FMT treatment could restore the GSH level but reduce the accumulation of MDA in hepatic tissues (**Fig 2c** and **2d**). Furthermore, a decreased alpha diversity was observed in the FMT group, similar to IR + Met group (**Fig. 2f**). Meanwhile, the attenuation of gut damage was showed by H&E staining, and the improvement in tight junction proteins ZO-1 and Occludin, and decreased HIRI-induced *IL-1 β* , *IL-6*, and *IL-18* (**Fig. 2g**–**2i**). Serum FD-4 level showed a sharply decrease in IR + FMT group (**Fig. 2j**). To further identify the function of gut microbes, experiments were designed, and combination treatment of antibiotics (1 mg/mL penicillin sulfate, 1 mg/mL neomycin sulfate, 1 mg/mL metronidazole and 0.16 mg/mL gentamicin) and metformin were employed for 1 week before IR treated. As results showed, the survival rate of IR + Met +

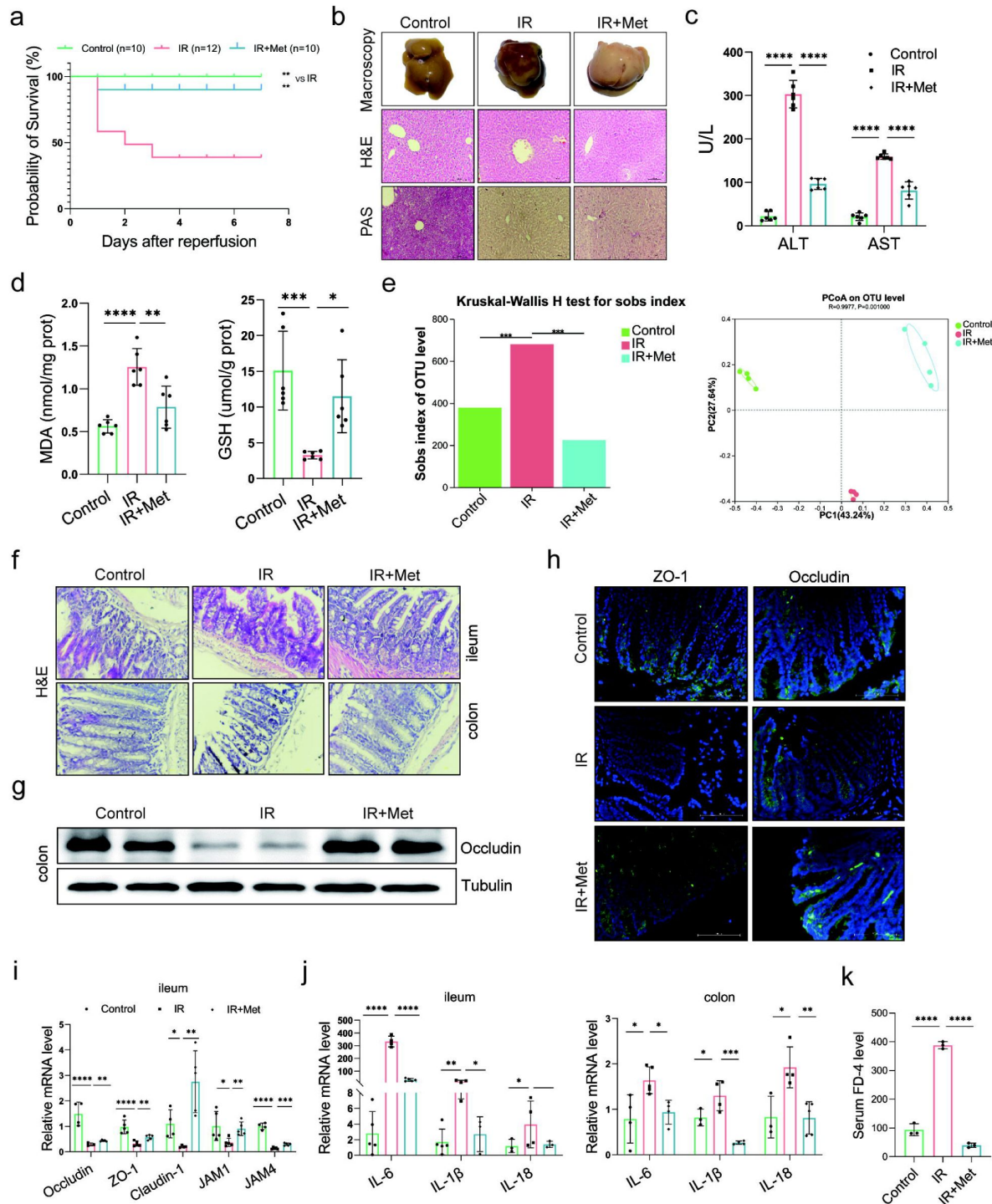


Figure 1.

Metformin significantly mitigates HIRI by reshaping gut microbiota in mice

(a) Survival rate analysis (n = 10-12/group), *Kaplan Meier* was used to analyze statistical differences (Control vs IR, and IR+Met vs IR). (b) Morphological appearance, H&E and PAS staining of liver (n = 3/group). (c) Serum ALT and AST levels. (d) Levels of MDA and GSH in liver tissue. (e) Sobs index, PCoA (n = 4/group). (f) H&E staining of ileum and colon (n = 3/group). (g) Western blot of Occludin in colon (n = 3/group). (h) IFC of ZO-1 and Occludin in colon (n = 3/group). (i) qRT-PCR analysis of *Occludin*, *ZO-1*, *Claudin-1*, *JAM1* and *JAM4* in ileum. (j) *IL-6*, *IL-1 β* and *IL-18* in ileum and colon. (n = 3-5/group). (k) FD-4 level in mice serum (n = 3). Data are expressed as mean $\pm$ standard deviation. *One-way ANOVA* was used to analyze statistical differences; **P* < 0.05, ***P* < 0.01 and ****P* < 0.001.

antibiotics (Abx) group was greatly reduced, while the levels of ALT, AST, MAD and GSH were completely opposite to IR + Met group (**Fig. 2k**). The results confirmed that gut microbiota is critical for the effect of metformin against HIRI.

FMT alleviates HIRI-induced ferroptosis through reshaped fecal microbiota

Since the reperfusion-induced excessive oxidative stress takes place in iron-rich hepatocytes,²⁶ experiments were designed to observe ferroptosis in HIRI. The ferroptosis inhibitor, DFO, significantly ameliorated the liver injury, suggesting that ferroptosis contributes to HIRI (sFig. 1). Hence, to find out the relation, ferroptosis-related indicators were also monitored in both IR + Met and IR + FMT groups. The increased accumulation of Fe after HIRI was significantly cut down by metformin and FMT treatment (**Fig. 3a**). The quantitative Real-time PCR (qRT-PCR) analysis showed that the HIRI-induced upregulation of ferroptotic parameters, *ACSL4*, *SLC7A11*, *SLC39A14* and *COX-2*, were significantly decreased by metformin and its reshaped gut microbiota, respectively (**Fig. 3b–3e**). The results of western blotting further confirmed the anti-ferroptosis of metformin and FMT by the remarkable reduction of *ACSL4*, *TFR1*, *VDAC1*, *VDAC2* and *VDAC3* levels and the apparent increase of anti-ferroptotic *FTTH1* and *Xct* levels (**Fig. 3f**). The results was further consolidated by lipid ROS staining and transmission electron microscopy (TEM) analysis (**Fig. 3g**). Additionally, transcriptome data on genes associated with anti-ferroptosis showed an increase in both IR + Met and IR + FMT groups, whereas proferroptotic genes showed a decrease, when compared with IR group (**Fig. 3h**). Collectively, the results suggested that FMT performed protection against HIRI by inhibiting ferroptosis.

Metformin induces GABA-producing gut microbiota

To elucidate the molecular mechanisms through which pathway participates metformin-treated IR injury, we analysed gene expression profiles of each group mice. Transcriptome sequencing analysis revealed that 9697 genes were in common among four groups (sFig. 6). Therefore, we used these common genes for KEGG analysis, showing that similar mRNA changes are mainly concentrated in the three top pathways: lipid metabolism, carbohydrate metabolism, and amino acid metabolism (**Fig 4a**). Given the close relevance between lipid metabolism and ferroptosis (sFig. 2), and the fact of carbohydrate metabolism is a primary way to metabolize amino acids, 22 species of amino acid were detected in liver tissues using HPLC-MS/MS for further identification of key metabolites involved in the role of metformin against HIRI-induced ferroptosis. It was found that only GABA level is significantly increased by metformin treatment and FMT treatment (sFig. 3), further verifying by the data of ELISA detection (**Fig. 4b**). Moreover, it was observed that the genus of *Bacteroides* had a significant increase based on the 16s rRNA gene sequencing of metformin-treated mice microbes (**Fig 4c**). qRT-PCR analysis of eukaryotic GABA synthesis key enzymes including *SSADH*, *GABA-T*, *Gad1* and *Gad2* in liver tissues, were not upregulated by metformin (**Fig. 4d**), which was also confirmed by western blotting detection of *GAD1* and *GAD2* in liver (**Fig. 4f**). Interestingly, the fecal GABA concentration was shown a significant increase in metformin-treated mice, which was copied by the IR + FMT group, but eliminated by antibiotics treatment (**Fig. 4e**). In order to further identify the increased GABA originates from gut microbiota, two key enzymes of prokaryotes GABA synthesis, *GAD* and *PAT*, were detected on DNA level, finding that both of them are significantly increased in the feces from IR+Met and IR+FMT groups (**Fig. 4g and h**). In order to clarify the specific effects of metformin on microbiota, given the big safety margin, healthy volunteers were recruited for a 1 week of daily oral 500mg dose of metformin trial. Fecal samples were collected before and after oral administration of metformin for metagenomic analysis. Detailedly, the species of *Bacteroides* containing *Bacteroides thetaiotaomicron*, *Bacteroides uniformis*, and *Bacteroides salyersiae*, were enriched in human gut after metformin administration (**Fig. 4i**). Excitingly, the increase of GABA production, and the expression of *GAD* and *PAT* were shown in the results of fecal experiments *in vitro* and *in vivo*,

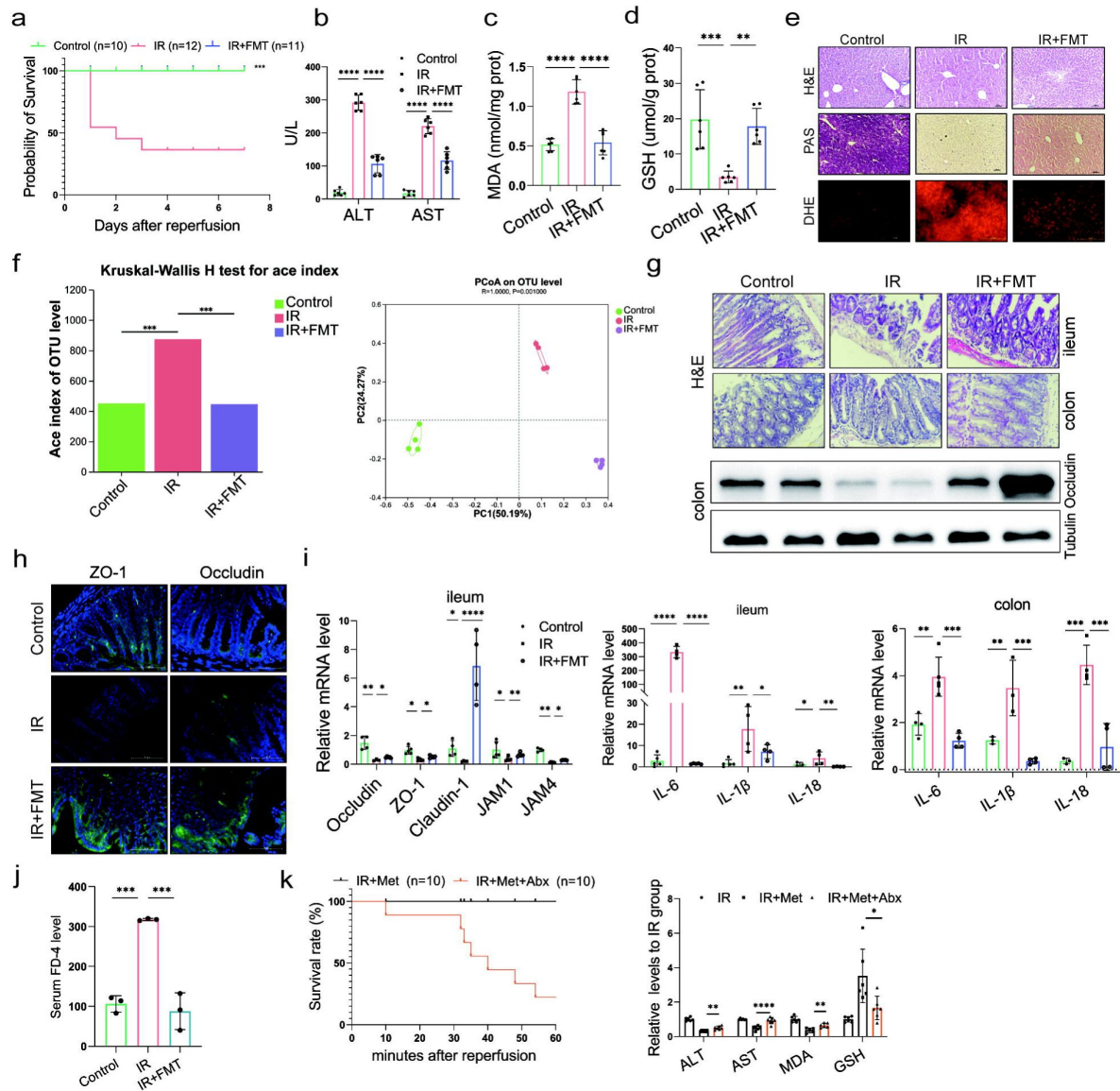


Figure 2.

Metformin-resaped fecal microbiota attenuates HIRI

(a) Survival rate analysis (n = 10-12/group), *Kaplan Meier* was used to analyze statistical differences. (b) Serum ALT and AST levels. Levels of MDA (c) and GSH (d) in liver tissue. (e) H&E, PAS and DHE staining of liver (n = 3/group). (f) Alpha diversity and PCoA level of mice microbes. (g) H&E staining of ileum and colon and Western blot analysis of Occludin in colon (n = 3/group). (h) IFC of ZO-1 and Occludin in colon (n = 3/group). (i) qRT-PCR analysis of *Occludin*, *ZO-1*, *Claudin-1*, *JAM1* and *JAM4* in ileum, and *IL-6*, *IL-1 β* and *IL-18* in ileum and colon. (n = 3-5/group). (j) FD-4 level in mice serum (n = 3). (k) Survival rate of IR + Met + Abx group (n = 10), *Kaplan Meier* was used to analyze statistical differences and the levels of serum ALT, AST, liver MDA, Fe, and GSH in IR + Met + Abx group. Data are expressed as mean $\pm$ standard deviation. *One-way ANOVA* was used to analyze statistical differences; * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$.

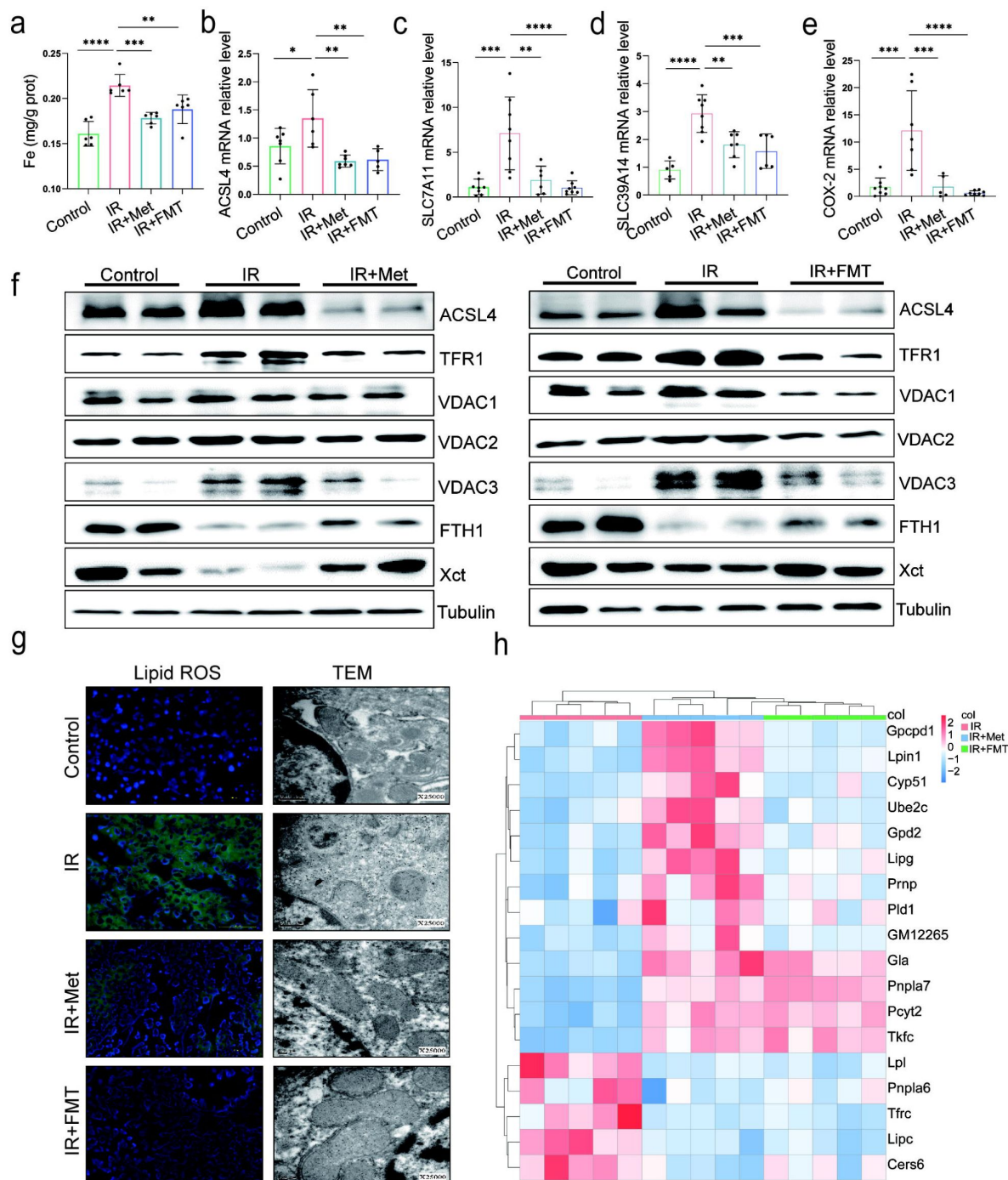


Figure 3.

Metformin alleviates HIRI through inhibiting ferroptosis

(a) Levels of Fe in liver tissue (n=6). qRT-PCR analysis of ACSL4 (b) *SLC7A11* (c), *SLC39A14* (d) and COX-2 (e) (n = 5-8/group). (f) Western blot analysis of ACSL4, FTH1, VDAC1, VDAC2, VDAC3, TFR1 and Xct in liver tissue (n = 3/group). (g) Lipid ROS staining and TEM analysis of liver tissues (n = 3/group). (h) Ferroptosis-related transcriptome cluster diagram heatmap. Data are expressed as mean $\pm$ standard deviation. One-way ANOVA was used to analyze statistical differences; * P < 0.05, ** P < 0.01 and *** P < 0.001.

accompanied by the decreased glutamic acid (Glu), a key substrate for gut microbiota to synthesize GABA (Fig. 4j and k). Besides, putrescine, another substrate of GABA synthesis, was not found in primary culture medium but appeared after 24h fermentation and then was consumed later in the anaerobic culture of human feces experiment. (Fig. 4l). Taken collectively, GABA-producing gut microbiota was upregulated by metformin.

GABA is the critical metabolite of metformin-reshaped gut microbiota against HIRI-induced ferroptosis

The animal experiment was designed to further observe the effects of GABA on HIRI-induced ferroptosis (Fig. 5a). As expected, the data showed that GABA treatment reduced histological damage and liver glycogen consumption (Fig. 5b). Serum liver function detection showed that ALT and AST activities were dramatically depressed after GABA treatment (Fig. 5c). The decreased MDA and Fe as well as increased GSH were observed in further determination, confirming the protection of GABA on liver (Fig. 5d). The results of western blotting showed the anti-ferroptosis effect of GABA treatment by the significant reduction in ACSL4, TFR1, VDAC1,2,3 and increase of anti-ferroptotic FTH1 levels, which had the same attenuating effect as metformin and FMT treatment (Fig. 5e). In addition, the clustergram showed GABA upregulated the genes of anti-ferroptosis and depressed the proferroptotic ones, which was similar to the effect after metformin treatment (Fig. 5f). In summary, GABA treatment reduced HIRI-induced ferroptosis.

Discussion

Metformin has been well researched for treating various liver diseases due to its multifarious effects. Notably, it has been well documented that oral administration of metformin leads to therapeutic effects on HIRI,²⁷ leaving a thought-provoking question in its underlying mechanism. In the current study, the changes in the gut microbiota altered by metformin are demonstrated to be necessary and sufficient for conferring liver injury protection. Moreover, it has been confirmed that metformin attenuated HIRI by the increased GABA from reshaped gut microbiota, through anti-ferroptosis.

Increasing evidence shows that the gut microbiota is critical for the pleiotropic actions of metformin.^{25, 28, 29} Interestingly, metformin was previously found to have a much higher concentration in the gut lumen than other organs,¹⁸ as also confirmed by the current study. In fact, that concentration favoured the action of metformin on reshaping gut microbes. Despite the distinct heterogeneity of metformin-reshaped gut microbiota in recent research, *Bacteroides* with a high abundance of gut microbes was reported to be reduced after metformin treatment in studies of hyperlipidemia and diabetes,^{19, 30} which other studies vary from for the different action of this bacteria.^{31, 32} In our study, we used a disease model of HIRI, which may have unique characteristics compared to other disease models. It is possible that the specific disease model influenced the response of the gut microbiota. Additionally, the starting microbiota of the recipients and the characteristics of the donor microbiota used for FMT could also play a role in the disparity. Our findings showed an increase in abundance of the genus of *Bacteroides* after metformin treatment supported by the data of mice and healthy volunteers. Detailedly, the increased *Bacteroides* especially *Bacteroides thetaiotaomicron*, *Bacteroides uniformis*, and *Bacteroides salyersiae* were observed in the human gut after metformin administration. The enriched *Bacteroides thetaiotaomicron*, a glutamate-fermenting commensal, was reported to be decreased in obese individuals and was inversely correlated with serum glutamate concentration.³³ Therefore, *Bacteroides thetaiotaomicron* might be the potential bacterium regulating GABA production. Little research has been done on *Bacteroides uniformis* and *Bacteroides salyersiae*, needing a further exploration of their characteristics in the future. Notably,

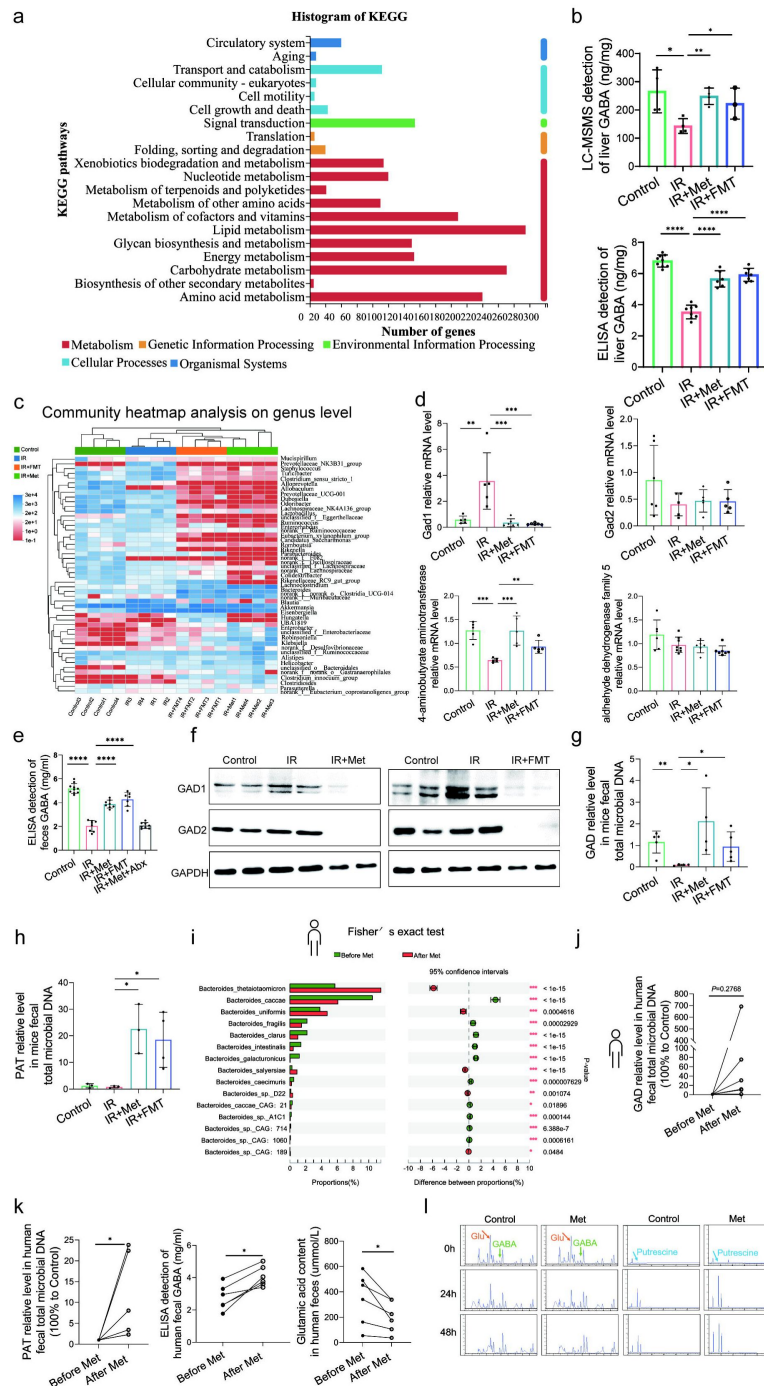


Figure 4.

Metformin induces GABA-producing gut microbiota.

(a) KEGG analysis of transcriptome. (b). HPLC-MS/MS of GABA content in liver, ELISA of GABA in liver (n = 6–9/group). (c) Community heatmap analysis on genus level. (d) qRT-PCR analysis of key metabolite enzyme glutamate decarboxylase 1 (Gad1), glutamic acid decarboxylase 2 (Gad2), 4-aminobutyrate aminotransferase (Gad3), aldehyde dehydrogenase family 5 (SSADH) in liver (n = 5–8/group). (e) ELISA of GABA in feces (n = 7–10/group). (f) Western-blot analysis of GAD1 and GAD2 in liver. (g–h) qRT-PCR analysis of GAD and PAT in mice feces (n = 3–5/group). (i) Metagenomic sequencing analysis of human feces. (j–k) The level of GAD, PAT, GABA and Glu in human feces (n = 5–6/group). (l) The level of GABA, Glu and putrescine in human feces vitro culture (n = 3/group). Data are expressed as mean ± standard deviation. One-way ANOVA was used to analyze statistical differences; **P* < 0.05, ***P* < 0.01 and ****P* < 0.001.

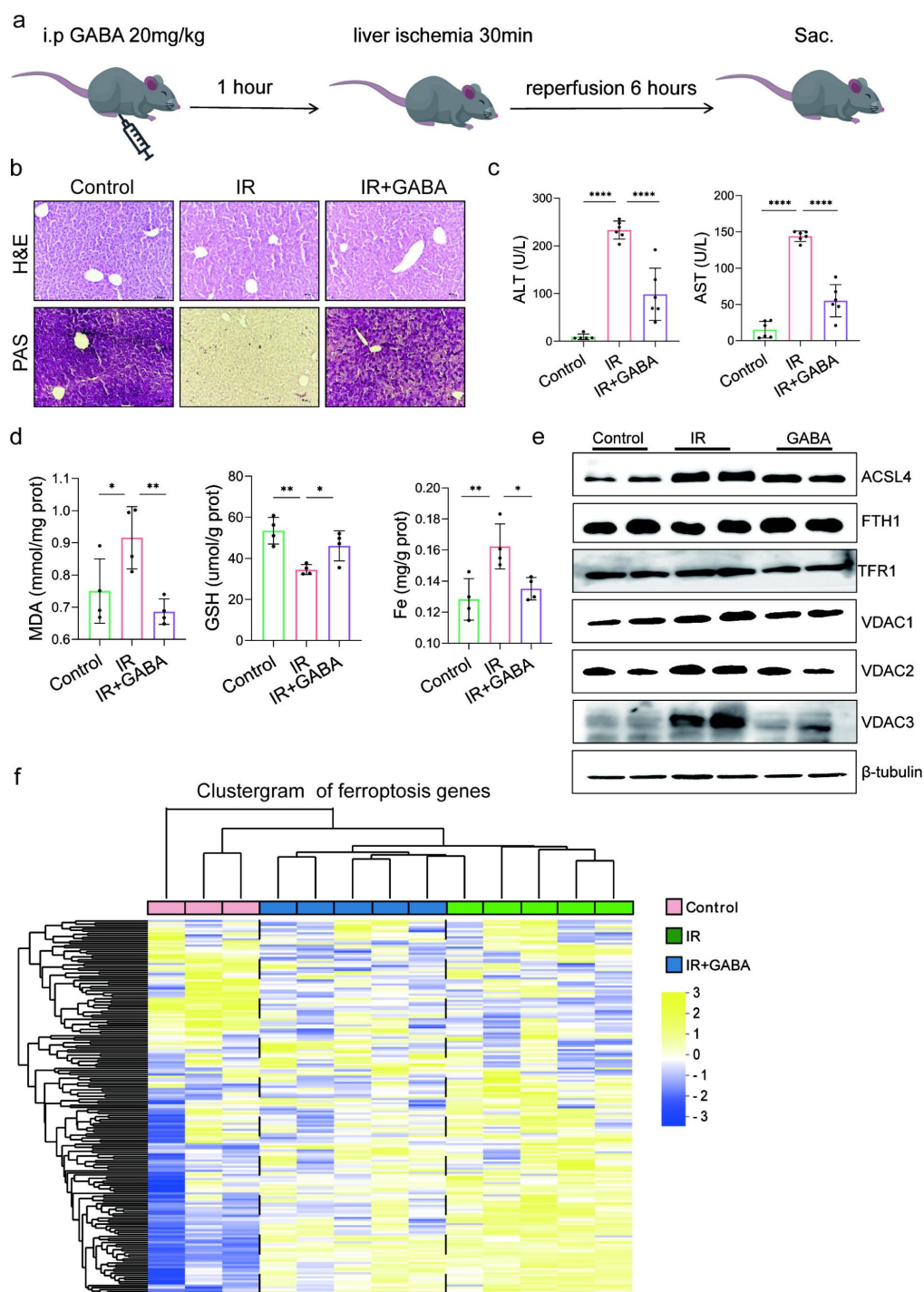


Figure 5.

GABA is the critical metabolite of metformin-reshaped gut microbiota against HIRI-induced ferroptosis.

(a) Diagram of GABA treatment experiment on mice model. (b) H&E and PAS staining of liver (n = 3/group). (c) Serum ALT and AST levels (n = 6/group). Levels of MDA, GSH and Fe in liver tissues (d) (n = 4/group). (e) Western blot analysis of ACSL4, FTH1, TFR1 and VDAC1,2,3 in liver (n = 4/group). (f) Ferroptosis-related transcriptome cluster diagram heatmap (n = 3-5/group). Data are expressed as mean ± standard deviation. One-way ANOVA was used to analyze statistical differences; * $P < 0.05$, ** $P < 0.01$ and *** $P < 0.001$.

Phascolarctobacterium, coexistly with *Bacteroides thetaiotaomicron*,³⁴ was reported as an emerging probiotic which showed a significant increase in our study. Taken collectively, those reshaped bacteria after metformin treatment may be responsible for attenuating HIRI.

Increased GABA was shown from the metformin reshaped gut microbiota for the first time, verified by our results of feces from mice and humans. As the critical metabolite of bacteria against acid stress, GABA was extensively produced by gut microbes, especially the species with strong capacity of acid resistance. Interestingly, our analysis of the sequences in PUBMED Genebank found that those three species of *Bacteroides*, *Bacteroides thetaiotaomicron*, *Bacteroides uniformis*, and *Bacteroides salyersiae*, which belong to metformin reshaped microbiota, contain the key enzyme genes of GABA synthesis GAD and PAT (sFig 4), further confirmed by the qRT-PCR result of human fecal DNA *in vivo* and *in vitro* experiment. Specifically, the species with the strong ability of producing GABA such as *Bacteroides thetaiotaomicron*, *Bacteroides uniformis* and *Bacteroides salyersiae* were increased in metformin-treated volunteers. Some studies were published to elucidate the mechanism of metformin regulating bile acids and folate metabolite by reshaped gut microbiota, however, there is no report about how metformin to affect GABA synthesis of gut microbes. The data from anaerobic cultured human feces showed that metformin promoted the increase of GAD and PAT, and GABA level along with the substrate consumption of glutamate and putrescine in a time-dependent manner.

GABA production was reported to be critical against acid stress for microbiota to survive and preserve metabolic activity, especially in the human gut with a lower pH than other animals.³⁵ Humans benefit from bacteria-derived GABA. Peripheral GABA was first demonstrated in intestinal flora in 1981 and further confirmed by recent studies.³⁶ In addition to the important role in the central neural system and immune system, GABA showed a protective effect in different acute liver injury models.^{37, 38} Published papers have shown that GABA performs beneficial effect through binding to the GABA receptors or promoting histone acetylation.^{39, 40} Our data from GABA-treated mice showed protective effects against HIRI-induced ferroptosis. However, further studies are required to examine the exact molecular mechanisms of GABA regulating ferroptosis.

In conclusion, we have demonstrated metformin-reshaped gut microbiota performs therapeutic responses for HIRI. Gut microbiota has been obviously modified while receiving oral metformin treatment, suggesting that metformin-reshaped gut microbiota has enhanced intestinal mucosal function so that reduce harmful substances to the liver, which most efficiently evade damage. Specifically, GABA producing bacteria were significantly regulated after metformin administration. Demonstrated for the first time, we showed that the increased GABA is a key metabolite of gut microbiota for metformin to mitigate HIRI. These results offer a unique therapeutic strategy, with optimized gut microbiota, that can serve as a target for metformin treating liver diseases.

Material and methods

Animals

Six-to-eight-week-old specific pathogen-free male C57BL/6 mice were purchased from Beijing Weitonglihua Experimental Animal Technology Co. Ltd. (Beijing, China). All mice were housed under controlled temperature and humidity conditions with a 12-h light-dark cycle and free access to food and water. The mice were fasted overnight before the experiments. All animal experiments were approved by the Institutional Animal Care and Use Committee of the Experimental Animal Centre of Wenzhou Medical University (wydw2021-0348).

Healthy volunteers

All research was conducted in accordance with both the Declarations of Helsinki and Istanbul, all research was approved by the Ethics Committee of the Second Affiliated Hospital of Wenzhou Medical University (2022-K-08-01), and written consent was given in writing by all subjects.

Hepatic ischemia/reperfusion (IR) mouse model

A mouse model for HIRI was established as previously described.³² Briefly, mice were anaesthetised with pentobarbital sodium. A noninvasive microvascular artery clip was placed on the left branches of the hepatic artery and portal vein for 30 min, and the clip was removed for reperfusion for 6 h. During the study period, body temperature was maintained at 37 °C using a heating lamp.

Experimental design

Mice were randomly divided into the several groups. In general, the arterial and portal vessels to the cephalad lobes were clamped for 30min and reperfusion for 6 h. No vascular occlusion in sham-controlled mice. In the treatment groups, animals were treated with metformin (1 mg/mL), metformin modified fecal microbiota transplantation or metformin (1 mg/mL) mixed with antibiotics (1 mg/mL penicillin sulfate, 1 mg/mL neomycin sulfate, 1 mg/mL metronidazole and 0.16 mg/mL gentamicin), in the drinking water for 1 week. Specifically, FMT-treated mice was fed with washed metformin-treated mice feces, which had been detected by HPLC showing that there was no metformin left in the feces. Other two groups of mice were infused at 1 h prior to the onset of liver ischemia with a single dose of deferoxamine (DFO) (20 mg/kg i.p.) or GABA (100 mg/kg i.p.), dissolved in PBS. Mice were sacrificed after reperfusion; liver and serum samples were collected for analysis.

Measurement of serum alanine transaminase (AST), aspartate aminotransferase (ALT) level, liver Fe content and GABA level

Serum AST and ALT levels were measured using assay kits (Nanjing Jiancheng Bioengineering Institute, C010-2-1, C009-2-1). Liver samples were homogenised in saline, and Fe concentrations were measured by tissue Fe assay kit (Nanjing Jiancheng Bioengineering Institute, A039-2-1). GABA level was detected by an ELISA kit (Wuhan MSKBIO, KT21124).

Measurement of lipid peroxidation, glutathione (GSH) and superoxide anion levels of liver tissues

Malondialdehyde (MDA) Detection Kit (Nanjing Jiancheng Bioengineering Institute, A003-1-2) was determined to select the MDA level as a marker of lipid peroxidation. The assay was performed according to the manufacturer's instructions. Lipid reactive oxygen species (ROS) was measured by BODIPY 581/591 C11 (MKbio, 217075-36-0). Sections were stained with BODIPY 581/591 C11 and nucleated with 4',6-diamidino-2-phenylindole (DAPI) (Beyotime Institute of Biotechnology, A0131). Images were acquired under a fluorescence microscope (Nikon, Tokyo, Japan). The liver was homogenised, and the supernatant was collected for GSH analysis using a GSH assay kit (Nanjing Jiancheng Bioengineering Institute, A006). Frozen sections of the liver (8Lµm) were placed on glass slides and incubated with 10Lmmol/L dihydroethidium (DHE) (Beyotime Institute of Biotechnology, S0063) in a dark container at 37 °C for 30Lmin. Sections were rinsed three times in phosphate-buffered saline (PBS) then observed using an upright microscope (Nikon, Tokyo, Japan).

Hematoxylin-eosin (H&E) and periodic acid-schiff (PAS) staining

For histopathological examination, liver and intestinal tissues were fixed in 4% paraformaldehyde overnight. Selected tissue blocks were processed using a routine overnight cycle in a tissue processor. Tissue blocks were then embedded in wax and serially sliced into 5-µm sections. H&E

and PAS staining performed as instructions provided by the reagent manufacturer (Solarbio, G1120, G1281). Visualization and images were obtained under an optical microscope (Nikon, Tokyo, Japan).

Histoimmunofluorescence staining

During immunofluorescence, sections were dyeing with primary antibody (1:100) Occludin (Proteintech, 27260-1-AP) and zonula occludens 1 (ZO-1) (Proteintech, 66378-1-IG) and incubated at 4°C overnight. Secondary antibody coupled with Alexa Fluor 488 (Bioss, bs-40296G-AF488) was incubated at 37°C for 1 h. After additional PBS washing, the sections were sealed with an antifluorescence quencher containing 4',6-diamidino-2-phenylindole (DAPI). Immunofluorescence images were obtained using a laser scanning confocal microscope (Nikon, Tokyo, Japan) and quantified by ImageJ (NIH, USA).

Transmission electron microscopy (TEM) analysis

After reperfusion, the mice were euthanized, and the livers were excised and washed with precooled PBS (pH 7.4). Part of the liver was then removed and incubated overnight in 0.1LM PBS (pH 7.4) containing 2.5% glutaraldehyde. The target tissues were cut into 50-μm-thick sections using a vibratome. Selected areas of the livers were postfixated in 1% osmium tetroxide for 1Lh, dehydrated in a graded ethanol series and embedded in epoxy resin. The polymerisation was performed at 80L°C for 24Lh. Ultrathin sections (100Lnm) were cut, stained with uranyl acetate and lead citrate and viewed under a JEM2000EX TEM (JEOL, Tokyo, Japan). Five fields were randomly selected for each sample to examine mitochondria with ferroptosis features.

Quantitative Real-time PCR (qRT-PCR) analysis

Total RNA was extracted from liver and intestinal tissues using TRIzol (Yamei, YY101). Isolated RNA was reverse transcribed to cDNA using a kit (Vazyme, R323-01). The obtained cDNA was subjected to PCR using primers designed to detect *Interleukin (IL)-6*, *IL-1β*, *IL-18*, *recombinant Solute Carrier Family 7, Member 11 (SLC7A11)*, *solute carrier family 39, member 14 (SLC39A14)*, *acyl-CoA synthetase long-chain family member 4 (ACSL4)*, *cyclooxygenase-2(COX-2)* *ZO-1*, *Occludin*, *Claudin-1*, *Junctional adhesion molecule (JAM)1*, *JAM4 succinic semialdehyde dehydrogenase (SSADH)*, *GABA transaminase (GABA-T)*, *Gad1 and Gad2* and β -actin (primer sequences are listed in sFig. 5). Gene expression was determined using the SYBR Green kit (Vazyme, Q711-02). All the results were normalised against β -actin expression using the Thermal Cycler Dice Real Time System (ABI QuantStudio6, Singapore).

Western blotting

Total protein samples were extracted from tissues using RIPA lysis buffer (Yamei, PC101). Protein concentrations were determined using a BCA protein detection kit (Yamei, ZJ101). Proteins were separated using 10% SDS-PAGE and transferred to polyvinylidene fluoride membranes. Membranes were blocked in 5% skimmed milk then incubated with the primary antibodies ACSL4 (AbCam, ab155282), Transferrin Receptor 1 (TFR1) (Abclonl, A5865), ferritin heavy chain (FTH1) (AbCam, ab65080), Occludin (Proteintech, 27260-1-AP), GAPDH (Proteintech, 0494-1-AP) and β -tubulin (Proteintech, 10068-1-AP) at 4 °C overnight. After washing three times with tris buffered saline, the membranes were incubated with appropriate anti-rabbit or anti-mouse secondary antibodies at room temperature for 1 h. Imprinting was observed using chemiluminescence (Yamei, SQ201) and an Odyssey imaging system (Li-Cor-Biosciences, NE, USA).

High-performance liquid chromatography with tandem mass spectrometric detection (HPLC–MS/MS) detection of metformin and liver amino acids

Sample separation was performed using a ZORBAX Eclipse XDB-C18 column (4.6 × 150 mm, Agilent, USA) using an injection volume of 5 µL. Metformin detection conditions are as follows: column temperature of 30°C, one mobile phase is NaH₂PO₄ (Sigma, V900060) containing 5 mmol/L sodium dodecyl sulfonate (Sigma, V900859) with pH 3.5 while the other mobile phase is acetonitrile (Aladdin, A104440). The metformin content in each sample was measured by HPLC (Agilent 1260, USA). Amino acid detection conditions are as follows: column temperature of 40°C, mobile phase A of 10% methanol/water (containing 0.1% formic acid), mobile phase B of 50% methanol/water (containing 0.1% formic acid) and flow rate of 0.4 mL/min. Mass spectrometry was performed using a triple quadrupole mass spectrometer with an ESI source in negative ionisation mode. High-purity nitrogen was used as the nebulising and drying gas. Quantification was performed in multiple reaction monitoring mode.

16S rRNA sequencing

16S rRNA amplicon sequencing was performed at Chunlab Inc. (Seoul, Korea) with MiSeq system (Illumina). Briefly, for preparation of MiSeq library amplicons, target gene (16S rRNA V3–V4 region) was amplified using 338F (5'-ACTCCTACGGGAGGCAGCAG-3') and 806R (5'-GGACTACHVGGGTWTCTAAT-3') primers, using an ABI GeneAmp 9700 PCR thermocycler (ABI, CA, USA). The PCR product was extracted from 2% agarose gel and purified using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA) according to the manufacturer's instructions and quantified using a Quantus Fluorometer (Promega, USA). Normalization was performed the counts of individual OTUs in a sample by dividing the total counts of all OTUs within that sample followed by a multiplication by resulting in relative abundance expressed.

Metagenomic sequencing

Total genomic DNA was extracted from human fecal samples as mentioned in 16S rRNA sequencing. DNA extract was fragmented to an average size of about 400 bp using Covaris M220 (Gene Company Limited, China) for paired-end library construction. Paired-end library was constructed using NEXTflex™ Rapid DNA-Seq (Bioo Scientific, Austin, TX, USA). Adapters containing the full complement of sequencing primer hybridization sites were ligated to the blunt-end of fragments. Paired-end sequencing was performed on Illumina NovaSeq/Hiseq Xten (Illumina Inc., San Diego, CA, USA) at Majorbio Bio-Pharm Technology Co., Ltd. (Shanghai, China) using NovaSeq Reagent Kits/Hiseq X Reagent Kits according to the manufacturer's instructions (www.illumina.com .

Statistical analysis

GraphPad Prism version 9.0 (GraphPad Software, San Diego, CA) was used for statistical treatment. Experimental data were shown as the mean ± SD. Two-tailed unpaired *Student's t-test* and *one-way ANOVA* with *Tukey's* correction were used for all comparisons of mice-related experiments. A *P* value < 0.05 was considered significant. The sample distribution was determined using a Kolmogorov–Smirnov normality test.

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Disclosure and competing interests statement

All the authors had no potential conflict of interest to declare.

Data availability

The original contributions presented in the study are publicly available. These data can be found here: <https://pan.quark.cn/s/aca2e6ffcbbc> 

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Harvard T.H. Chan School of Public Health, Boston, United States of America

Reviewer #1 (Public Review):

The major strength of this work is its scope, including detailed mouse phenotyping, interdisciplinary methods, and numerous complementary experiments. The antibiotic depletion and FMT experiments provide support for a role of the gut microbiota in this mouse model.

A major limitation is the lack of studies narrowing down which microbes are responsible. Sequencing data is shown, but no follow-up studies are done with bacterial isolates or defined communities.

The link to GABA is also somewhat tenuous. While it does match the phenotypic data, there are no targeted experiments in which GABA producing microbial communities/strains are compared to a control community/strain. As such, it seems difficult to know how much of the effects in this model are due to GABA vs. other metabolites.

My major recommendation would be to revise the title, abstract, and discussion to provide more qualification and to consider alternative interpretations.

Some key controls are also missing, which could be addressed by repeat experiments in the mouse model. The antibiotic depletion experiment would be improved by testing the effect of antibiotics in the absence of metformin, to see if the effect is just driven by the model itself as opposed to an interaction between metformin and antibiotics. The FMT experiment lacks a control group and suffers from pseudoreplication: multiple donors from metformin treated and untreated mice could be used to colonize separate groups of recipient mice.

- <https://doi.org/10.7554/eLife.89045.2.sa1>

Reviewer #2 (Public Review):

The authors examine the use of metformin in the treatment of hepatic ischemia/reperfusion injury (HIRI) and suggest the mechanism of action is mediated in part by the gut microbiota and changes in hepatic ferroptosis. While the concept is intriguing, the experimental approaches are inadequate to support these conclusions.

The histological and imaging studies were considered a strength and reveal a significant impact of metformin post-HIRI.

Weaknesses largely stem from the experimental design. The impact of metformin on the microbiota is profound resulting in changes in bile acid, lipid, and glucose homeostasis. Throughout the manuscript no comparisons are made with metformin alone which would better capture the metformin-specific effects. With the pathology and metabolic disturbances resulting from HIRI, it is important to understand if metformin is providing beneficial effects from reported mechanisms such as changes in bile acid, glucose, and/or lipid metabolism, or are these changes the result of a new unappreciated mechanism. A comparison of the reported and the new pathways is not included.

Overall, while the concept is interesting and has potential to better understand the pleiotropic functions of metformin, the limitations with the experimental design and lack of key controls make it challenging to support the conclusions.

- <https://doi.org/10.7554/eLife.89045.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Many drugs have off-target effects on the gut microbiota but the downstream consequences for drug efficacy and side effect profiles remain unclear. Herein, Wang et al. use a mouse model of liver injury coupled to antibiotic and microbiota transplantation experiments. Their results suggest that metformin-induced shifts in gut microbial community structure and metabolite levels may contribute to drug efficacy.

This study provides valuable mechanistic insights that could be dissected further in future studies, including efforts to identify which specific bacterial species, genes, and metabolites play a causal role in drug response. Importantly, although some pilot data from human subjects is shown, the clinical relevance of these findings for liver disease remain to be determined.

Thank you for reviewing our manuscript. We appreciate your valuable feedback. We agree that the downstream consequences of off-target effects on the gut microbiota by various drugs remain unclear. Our study aimed to shed light on this aspect by utilizing a mouse model of liver injury and conducting antibiotic and microbiota transplantation experiments. Our findings suggest that shifts in the structure and metabolite levels of the gut microbial community induced by metformin play a role in the drug's efficacy. We believe that these mechanistic insights provide a strong foundation for further investigations. Specifically, future studies could focus on identifying the specific bacterial species, genes, and metabolites that have a causal role in drug response. While we have included some pilot data from human subjects, we acknowledge that the clinical relevance of our findings in the context of liver disease still requires further determination. In fact, we focused on the alteration of microbiota and metabolism caused by metformin in human bodies, which could capture the characteristics of changes in a more composite clinical direction, elucidating the potential role of metformin. We appreciate your attention to this aspect and thank you again for your thoughtful review and valuable suggestions.

The major strength of this work is its scope, including detailed mouse phenotyping, interdisciplinary methods, and numerous complementary experiments. The antibiotic depletion and FMT experiments provide support for a role of the gut microbiota in this mouse model.

A major limitation is the lack of studies narrowing down which microbes are responsible. Sequencing data is shown, but no follow-up studies are done with bacterial isolates or defined communities.

We acknowledge the limitation of our study in not narrowing down the specific microbes responsible for the observed effects. We hold the opinion that metformin exerts its effects through modulation of specific metabolic pathways unique to the microbial community. Previous study has shown that metformin can inhibit microbial folate metabolism, leading to longevity-promoting effects that are not attributed to a single colony or strain[1]. Similarly, the impact of metformin on amino acid metabolism in the microbial community appears to be widespread. While further investigations with bacterial isolates or defined communities are needed, our findings suggest that metformin's effects on microbial metabolism are complex and involve multiple members of the microbial community.

The link to GABA is also somewhat tenuous. While it does match the phenotypic data, there are no targeted experiments in which GABA producing microbial communities/strains are compared to a control community/strain. As such, it seems difficult to know how much of the effects in this model are due to GABA vs. other metabolites.

We agree with your point regarding the tenuous link to GABA in our study. While we did observe an increase in GABA as the only amino acid following metformin treatment, and this finding has not been reported previously, we acknowledge the need for targeted experiments comparing GABA-producing microbial communities/strains to control communities/strains. Previous literatures suggest that metformin's modulation of the microbiota can vary significantly depending on the disease context, with different microbial populations exhibiting differential responses[2-4]. Given this complexity, we opted to study the overall

microbial community response to metformin rather than focusing on specific strains. Additionally, our detection of key enzymes involved in GABA synthesis at the community level further supports our findings.

My major recommendation would be to revise the title, abstract, and discussion to provide more qualification and to consider alternative interpretations.

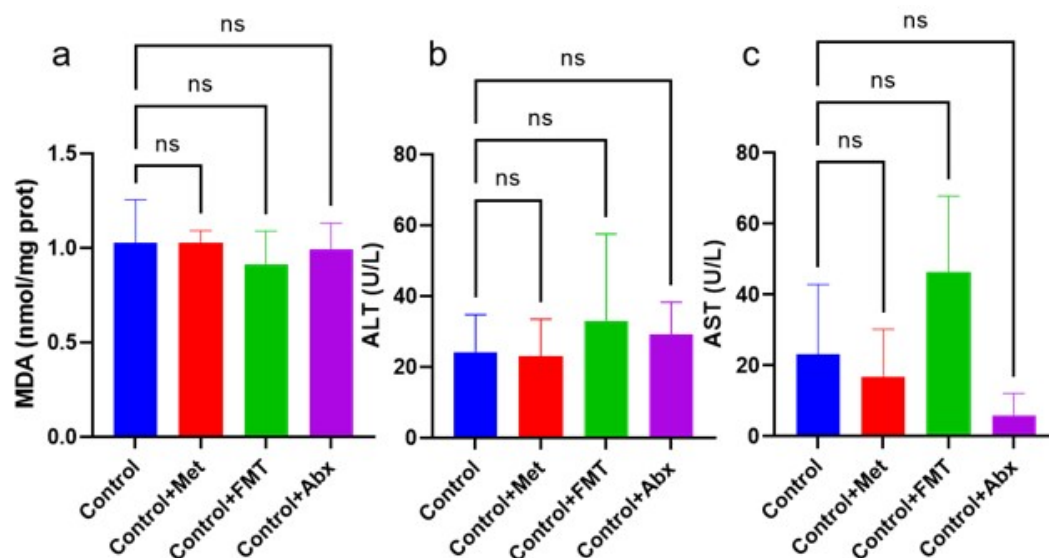
We appreciate your feedback and understand your concern regarding the need for more qualification and consideration of alternative interpretations. We hope to have more specific and detailed suggestions you may have to enhance the clarity and qualification of our title and abstract. Furthermore, we have tried to revise discussion in order to enhance the scientific rigor and logical coherence of our study. If you have any specific recommendations or insights, we would be more than willing to make further revisions to address those concerns.

Some key controls are also missing, which could be addressed by repeat experiments in the mouse model.

We appreciate your suggestion to include additional key controls in the mouse model experiments. We have conducted repeat experiments to test the effect of antibiotics in the absence of metformin to differentiate between the effects of the model itself and the interaction of metformin with antibiotics. As results of liver injury indicators shown, there were no significance among Control, Control+Met, Control+FMT and Control+Abx groups, revealing that metformin and its treated feces, and antibiotics had no effect on liver function in normal mice (Figure 1).

Author response image 1.

Figure1 a: Liver MDA detection; b: Serum ALT level; c: Serum AST level.



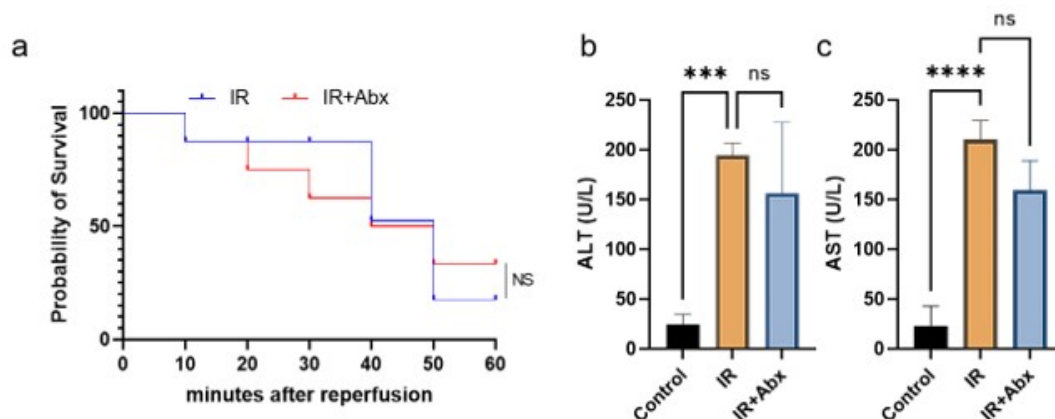
The antibiotic depletion experiment would be improved by testing the effect of antibiotics in the absence of metformin, to see if the effect is just driven by the model itself as opposed to an interaction between metformin and antibiotics.

For the antibiotic depletion experiment, we had used antibiotics (Abx) for the mice of modeling, and the survival rate and liver function detection suggested that Abx had no extra

effect on liver, which demonstrated that the effect is just driven by the model itself as opposed to an interaction between metformin and antibiotics (Figure 2).

Author response image 2.

Figure2 a: Survival rate between IR and IR + Abx group; b: Serum ALT level; c: Serum AST level.



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Reviewer #2 (Public Review):

The authors examine the use of metformin in the treatment of hepatic ischemia/reperfusion injury (HIRI) and suggest the mechanism of action is mediated in part by the gut microbiota and changes in hepatic ferroptosis. While the concept is intriguing, the experimental approaches are inadequate to support these conclusions.

The histological and imaging studies were considered a strength and reveal a significant impact of metformin post-HIRI.

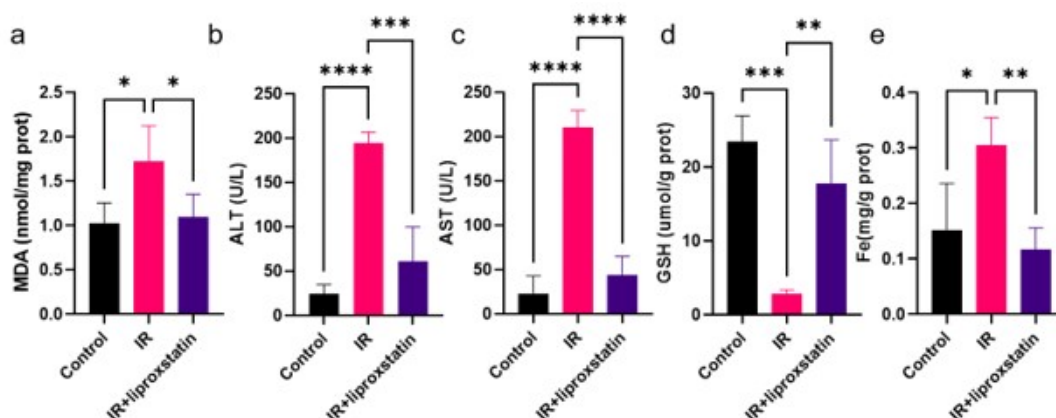
Thank you for reviewing our paper titled “Gut microbiota-derived gamma-aminobutyric acid from metformin treatment reduces hepatic ischemia/reperfusion injury through inhibiting ferroptosis”. We appreciate your insightful comments and suggestions, which have provided valuable insights into improving the quality and credibility of my research. We agree with your assessment that the experimental approaches used in this study may have limitations in supporting the conclusions drawn, and we appreciate your recognition of the strength of our histological and imaging studies, which clearly demonstrate the impact of metformin post-HIRI.

Weaknesses largely stem from the experimental design. First, use of the iron chelator DFO would be strengthened using the ferroptosis inhibitor, liproxstatin.

Your suggestion to employ the ferroptosis inhibitor, liproxstatin, in addition to the iron chelator DFO is well-taken. Incorporating liproxstatin into our experimental setup would provide a more comprehensive understanding of the involvement of hepatic ferroptosis in the mechanism of action of metformin. Therefore, we employed liproxstatin to inhibit HIRI and detected some core indicators of liver injury. As figure 3 shown, liproxstatin can reduce liver injury, restore liver GSH level and inhibit Fe accumulation, suggesting that ferroptosis plays an important role in HIRI. We hope this modification will enhance the credibility of our conclusions.

Author response image 3.

Figure3 a: Liver MDA detection; b: Serum ALT level; c: Serum AST level; d: Liver GSH level; e: Liver Fe level.



Second, the impact of metformin on the microbiota is profound resulting in changes in bile acid, lipid, and glucose homeostasis. Throughout the manuscript no comparisons are made with metformin alone which would better capture the metformin-specific effects.

Thank you for raising an important point regarding the impact of metformin on the microbiota and its potential effects on bile acid, lipid, and glucose homeostasis. It has well known that that the effects of metformin on normal blood glucose and lipid metabolism are minimal. Metformin primarily exerts its effects in cases of impaired glucose tolerance, which is why it is widely used for non-diabetic conditions. Regarding the changes in bile acid metabolism and chronic cholesterol and lipid elevation, these associations are typically observed in chronic liver disease models. Since our study focuses on an acute model of HIRI, we did not specifically investigate these changes.

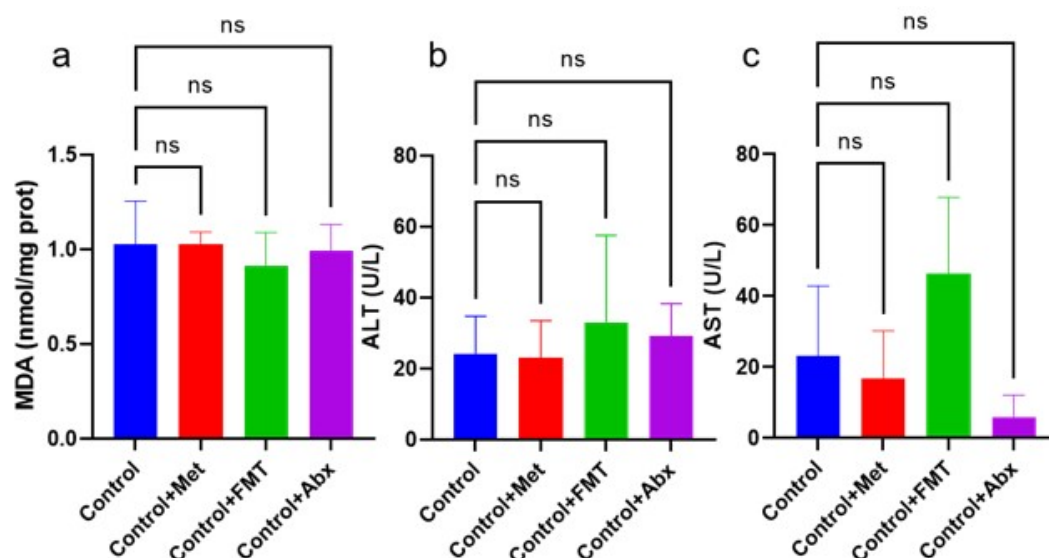
Lastly, the absence of proper controls including germ free mice, metformin treated mice, FMT treated mice, etc make it difficult to understand the outcomes and to properly reproduce the findings in other labs.

Lastly, we acknowledge your concern regarding the absence of proper controls, including germ-free mice, metformin-treated mice, and FMT -treated mice. We understand that these controls are essential for robustly interpreting and reproducing our findings. Therefore, we have added a batch of experiments for verification. As results shown, there were no

significance among Control, Control+Met, Control+FMT and Control+Abx groups, revealing that metformin and its treated feces, and antibiotics had no effect on liver function in normal mice (Figure 1). We hope the result of these controls could address your valid point and provide a more comprehensive framework for understanding the outcomes.

Author response image 4.

Figure1 a: Liver MDA detection; b: Serum ALT level; c: Serum AST level.



Overall, while the concept is interesting and has the potential to better understand the pleiotropic functions of metformin, the limitations with the experimental design and lack of key controls make it challenging to support the conclusions.

We genuinely appreciate your constructive criticism and the time you have taken to evaluate my work. Your feedback has shed light on the limitations of our experimental design and the need for key controls, which we have addressed in revised manuscript. If you have any further recommendations or concerns, we would be more than willing to incorporate them into my future work.

Reviewer #3 (Public Review):

The study presented in this paper explores the role of gut microbiota in the therapeutic effect of metformin on HIRI, as supported by fecal microbiota transplantation (FMT) experiments. Through high throughput sequencing and HPLC-MS/MS, the authors have successfully demonstrated that metformin administration leads to an increase in GABA-producing bacteria. Moreover, the study provides compelling evidence for the beneficial impact of GABA on HIRI.

Thank you for your valuable feedback on our paper exploring the role of gut microbiota in the therapeutic effect of metformin on hepatic ischemia-reperfusion injury (HIRI). We appreciate your positive remarks and suggestions for improvement. In response to your comments, we have revised the manuscript accordingly. We have included additional details on the high throughput sequencing and HPLC-MS/MS methods used to analyze the gut microbiota and GABA levels. This should provide readers with a clearer understanding of our experimental approach and the evidence supporting our findings.

Regarding your suggestion to further investigate the mechanisms underlying the beneficial impact of GABA on HIRI, we agree that this is an important direction for future research. We plan to conduct additional studies to explore the specific mechanisms by which GABA exerts its protective effects on HIRI in the future. We also supplemented discussion of potential therapeutic strategies targeting GABAergic pathways in the discussion section.

Thank you once again for your insightful comments. We believe that these revisions have strengthened the manuscript and improved its scientific rigor. We hope that you find the revised version to be satisfactory and look forward to your further feedback.

Reviewer #1 (Recommendations For The Authors):

The writing could be improved. Multiple typos are found throughout and there is an overuse of adverbs like "expectedly". You should let the reader decide what is or is not expected. Try to avoid terms like "confirmed" or "validated", which only applies if you knew the result a priori. Remove underscores in species names. The Results section is also very difficult to interpret given the lack of explanation of experimental design. For example, the human study is only briefly mentioned within a larger paragraph on mouse data, without any explanation as to the study design. Similar issues are true for the transcriptomics and amplicon sequencing - it would help the reader to explain what samples were processed, the timepoints, etc.

Thank you for your valuable feedback on our manuscript entitled “Gut microbiota-derived gamma-aminobutyric acid from metformin treatment reduces hepatic ischemia/reperfusion injury through inhibiting ferroptosis” We appreciate your constructive comments and insightful suggestions for improvement.

We have carefully reviewed your comments and have made several revisions to enhance the clarity and readability of the manuscript. We have addressed the issue of multiple typos and have removed the overuse of adverbs, such as “expectedly,” to allow readers to draw their own conclusions from the results. Additionally, we have eliminated terms like “confirmed” or “validated” that may imply a priori knowledge of the results.

We apologize for the lack of clarity regarding the experimental design in the Results section. We have now provided a more detailed explanation of the study design for the human study, transcriptomics, and amplicon sequencing experiments. This includes information on the samples processed, timepoints, and other relevant details, to aid readers in understanding the experimental procedures.

In response to your comment about removing underscores in species names, we have revised the text accordingly to ensure consistency and accuracy in the species nomenclature used throughout the manuscript.

Once again, we sincerely appreciate your valuable input, which has helped us improve the quality of our manuscript. We hope that the revised version now meets your expectations and look forward to any further feedback you may have.

Thank you for your time and attention.

| *Line 53 - prebiotics aren't "microbial agents"*

We apologize for this error, which we have corrected. (line 55: “Microbial agents, such as synbiotics, prebiotics and probiotics...”)

| *Line 88 - sequencing doesn't "verify the critical role of gut microbiota"*

We apologize for this error, which we have corrected. (line 90: “In order to verifyclarify the critical role of gut microbiota in the pleiotropic actions of metformin,22-24 fecal samples were collected from the mice to perform 16S rRNA sequencing.

| *Line 92 - missing a citation for the "microbiota-gut-liver axis theory"*

We have corrected it in manuscript. (line 93: “Next, as the microbiota-gut-liver axis theory indicates,25 HIRI-induced dysfunction of the gut barrier may aggravate liver damage by disrupting the gut microbiota.”)

| *Line 112 - it's very surprising to me that FMT led to lower alpha diversity, which seems impossible.*

We understand your surprise regarding the observed decrease in alpha diversity after FMT. Our findings indeed deviate from the commonly observed pattern of increased alpha diversity post-FMT. We have carefully re-examined our data and conducted additional analyses to ensure the accuracy of our results. After thorough investigation, we have identified a potential reason for this unexpected outcome, which we believe could shed light on this phenomenon. We hypothesize that the lower alpha diversity observed in our study might be attributed to the specific characteristics of the donor microbiota used for FMT. While the donor microbiota exhibited certain beneficial properties associated with the therapeutic effect on HIRI, it could have presented a limited diversity compared to the recipient's original gut microbiota. This discrepancy in diversity could have contributed to the observed decrease in alpha diversity following FMT.

To further support our hypothesis, we have included a discussion on this unexpected finding in the revised manuscript. We believe that this addition will provide a more comprehensive understanding of the results and help contextualize the observed decrease in alpha diversity following FMT.

| *Line 117 - Antibiotics don't "identify the function of gut microbes." Need to specify which antibiotics were used and for how long.*

We have corrected it in manuscript. (line 119: “To further identify the function of gut microbes, experiments were designed, and combination treatment of antibiotics (1 mg/mL penicillin sulfate, 1 mg/mL neomycin sulfate, 1 mg/mL metronidazole and 0.16 mg/mL gentamicin) and metformin were employed for 1 week before IR treated.”)

| *Line 120 - this experiment shows that the gut microbiota (or antibiotics more precisely) matters, not the "reshaped gut microbiota"*

We have corrected it in manuscript. (line 124: “The results confirmed that reshaped gut microbiota is critical for the effect of metformin against HIRI.”)

| *Line 122 - need to reword this subheading and the concluding sentence. The main takeaway is that the FMT improved markers of ferroptosis, but no additional causal links are provided here.*

We have revised in manuscript. (line 125: “FMT alleviates HIRI-induced ferroptosis through reshaped fecal microbiota.”)

| *Line 141 - need to explain what transcriptomics data was generated and how it was analyzed.*

We have revised in manuscript. (line 144: “To elucidate the molecular mechanisms through which pathway participates metformin-treated IR injury, we analysed gene expression profiles of each group mice. Transcriptome sequencing analysis revealed that 9697 genes were in common among four groups (Supplementary Figure 6). Therefore, we used these common genes for KEGG analysis, showing that The transcriptome analysis of liver tissues showed that similar mRNA changes between Met group and FMT group are mainly concentrated in the three top pathways: lipid metabolism, carbohydrate metabolism, and amino acid metabolism (Fig 4a).”)

| *Line 150 - change to "16S rRNA gene sequencing". Typo: "mice microbes".*

We have revised in manuscript. (line 156: “Moreover, it was observed that the genus of Bacteroides had a significant increase based on the 16s rRNA gene sequencing of metformin-treated mice microbes.”)

| *Line 152 - upregulated refers to gene expression, change to enriched.*

We have revised in manuscript. (line 171: “Detailedly, the species of Bacteroides containing Bacteroides thetaiotaomicron, Bacteroides uniformis, and Bacteroides salyersiae, were enriched in human gut after metformin administration (Fig. 4i).”)

| *Line 159 - typo: "prokaryotes"*

We have revised in manuscript. (line 165: “In order to further identify the increased GABA originates from gut microbiota, two key enzymes of prokaryotes protokaryotic GABA synthesis, GAD and PAT, were detected on DNA level, finding that both of them are significantly increased in the feces from IR+Met and IR+FMT groups (Fig. 4h).”)

| *Line 161 - the human study should be under a new sub-heading and provide more details.*

We have revised in manuscript. (line 168: In order to clarify the specific effects of metformin on microbiota, given the big safety margin, healthy volunteers were recruited for a 1 week of daily oral 500mg dose of metformin trial. Fecal samples were collected before and after oral administration of metformin for metagenomic analysis .”)

| *Line 197 - It's unclear why the current study conflicts with prior literature. Is it due to the disease model, the starting microbiota, something else? Please add more discussion.*

Thank you for bringing this important point to our attention, and we appreciate your valuable input. We agree that it is important to discuss the potential reasons for the discrepancy between our findings and prior literature on metformin-reshaped microbiota. In our study, we used a disease model of HIRI, which may have unique characteristics compared to other disease models. It is possible that the specific disease model influenced the response of the gut microbiota. Additionally, the starting microbiota of the recipients and the characteristics of the donor microbiota used for FMT could also play a role in the disparity. We have expanded the discussion section of our revised manuscript to further address these potential factors and their implications. We hope that this additional information will provide a more comprehensive explanation for the discrepancy between our study and prior literature.

| *Figure 1a - change to Kaplan Meier not ANOVA. Specify the contrast - which groups are being compared?*

We have revised in Figure 1a.

Figure 1e, alpha diversity - relabel "sobs" with "observed OTUs". Change to 3 bars with error and add statistics.

We have revised in Figure 1e.

Figure 1e, PCA - this should be a separate panel (1f). Color of big red circle doesn't match the points. Add PERMANOVA p-value/R². Change to OTUs not genera. Better yet, use amplicon sequence variants from DADA2.

We have revised in Figure 1e..

Figure 2a - Change to Kaplan Meier. Also, it's unclear if residual metformin could be in the donor samples.

We have revised in Figure 2a.

Figure 2f, alpha diversity - relabel "sobs" with "observed OTUs". Change to 3 bars with error and add statistics.

We have revised in Figure 2f.

Figure 2f, PCA - this should be a separate panel (2g). Color of big orange circle doesn't match the points. Add PERMANOVA p-value/R². Change to OTUs not genera. Better yet, use amplicon sequence variants from DADA2.

We have revised in Figure 2f.

Figure 4b - check units, shouldn't this be ng/mg (i.e. weight not volume).

We have revised in Figure 4b.

Figure 4c,d - need more explanation in the legend and Results as to what is shown here.

We have revised in Figure 4c,d.

Figure 4d - unclear why only Bacteroides are shown here or if the p-values are adjusted for multiple comparisons.

Thank you for your comment regarding Figure 4d in our manuscript. We apologize for the confusion caused. The reason why only Bacteroides is shown in Figure 4d is because we specifically wanted to investigate the changes in Bacteroides abundance following metformin treatment.

In the mouse experiments, we observed a significant increase in Bacteroides after metformin treatment. To investigate if a similar change occurs in healthy volunteers, we examined the levels of Bacteroides in fecal samples before and after oral administration of metformin. We found that the abundance of Bacteroides also increased in the human gut after metformin administration, consistent with the results from the animal experiments. Regarding the p-values, we apologize for not mentioning whether they were adjusted for multiple comparisons in the figure legend. In our revised manuscript, we have provided a clarification stating that the p-values were adjusted using the appropriate method. We appreciate your

feedback and hope that this explanation clarifies the rationale behind Figure 4d. Thank you for your valuable input.

Reviewer #2 (Recommendations For The Authors):

Below I've listed several suggestions to improve the paper.

1. Controls - the authors should include metformin only treated mice, FMT only treated mice, etc. Additionally, germ free mice treated with metformin and HIRI would be helpful to better implicate the gut microbiome in these beneficial effects.

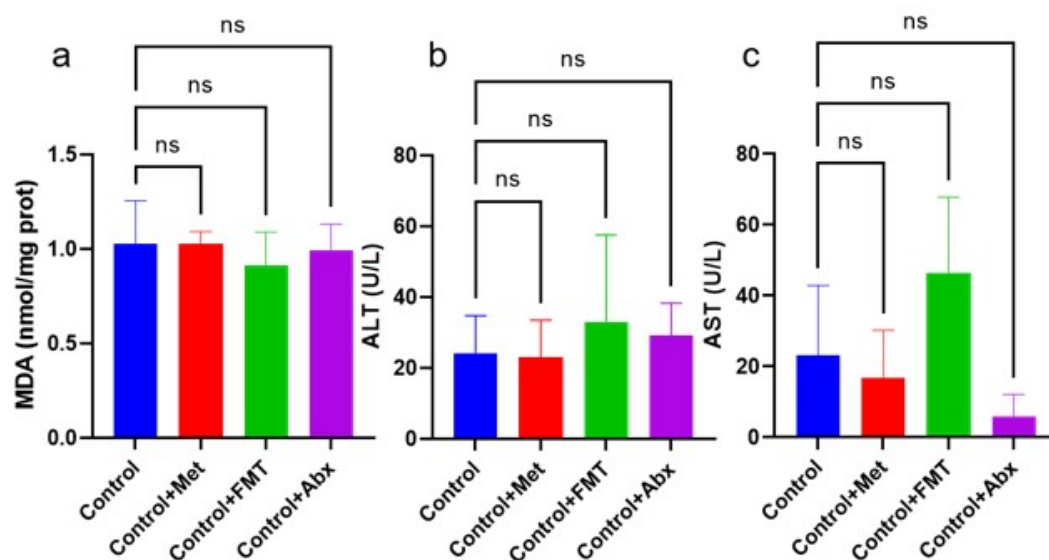
Thank you for your suggestion regarding the inclusion of additional control groups in our study. We agree that including metformin only treated mice, FMT only treated mice, and germ-free mice treated with metformin and HIRI would provide valuable insights into the role of the gut microbiome in the observed beneficial effects.

Therefore, we have included metformin only treated mice, FMT only treated mice and Abx only treated mice as supplement to better assess the specific contribution to the observed effects. As results shown, there were no significance among Control, Control+Met, Control+FMT and Control+Abx groups, revealing that metformin and its treated feces, and antibiotics had no effect on liver function in normal mice (figure1).

We appreciate your input and believe that the inclusion of these additional control groups will strengthen our study and provide a more comprehensive understanding of the role of the gut microbiome in the therapeutic effects observed.

Author response image 5.

Figure1 a: Liver MDA detection; b: Serum ALT level; c: Serum AST level.



1. More thorough characterization of metabolite pools. Metformin is known to influence many pathways including bile acids and lipids. These important molecules should be measures as they likely play a key role in the observed protective effect. In fact, many of the key changes displayed in Figure 3H are involved in lipid metabolism.

Thank you for your valuable feedback regarding the characterization of metabolite pools in our study. We appreciate your suggestion to measure the influence of metformin on bile acids and lipid metabolism, as they are crucial pathways that may play a significant role in the observed protective effect.

Regarding bile acids, we agree that they are important in the context of metformin's influence on metabolic pathways. However, it is important to note that the impact of metformin on bile acids appears to be more prominent in chronic liver disease models. In our acute model, the changes in bile acids were not as significant. Instead, our results primarily indicate a close association between lipid changes and hepatic ferroptosis. Metformin significantly modulates lipid metabolism, thereby alleviating liver ferroptosis.

Additionally, we have conducted metagenomic sequencing on the gut microbiota of healthy volunteers before and after oral administration of metformin. While analyzing the data, we did not observe significant changes in key genes involved in regulating bile acid variations. This might be attributed to the healthy volunteers used in our study, where significant changes in bile acids were not induced.

We appreciate your insightful comments and suggestions, which have shed light on the importance of characterizing bile acids and lipid metabolism in our study. While the impact of bile acids may be more evident in chronic liver disease models, our findings highlight the significant influence of metformin on lipid metabolism, closely related to hepatic ferroptosis. We will take your suggestions into account for future studies to further explore the role of bile acids and their regulation by metformin.

1. Imaging of lipid ROS is not quantitative. The authors should conduct more standard assays with BODIPY 581/591 C11 using cell lysates.

We appreciate your suggestion to conduct more standard assays using BODIPY 581/591 C11 with cell lysates.

We would like to clarify that we did indeed utilize assays with BODIPY 581/591 C11 to detect and measure lipid ROS in our study. The detailed description of these assays can be found in the Methods section of our paper. We followed established protocols and guidelines to ensure accurate and reliable measurements of lipid ROS levels.

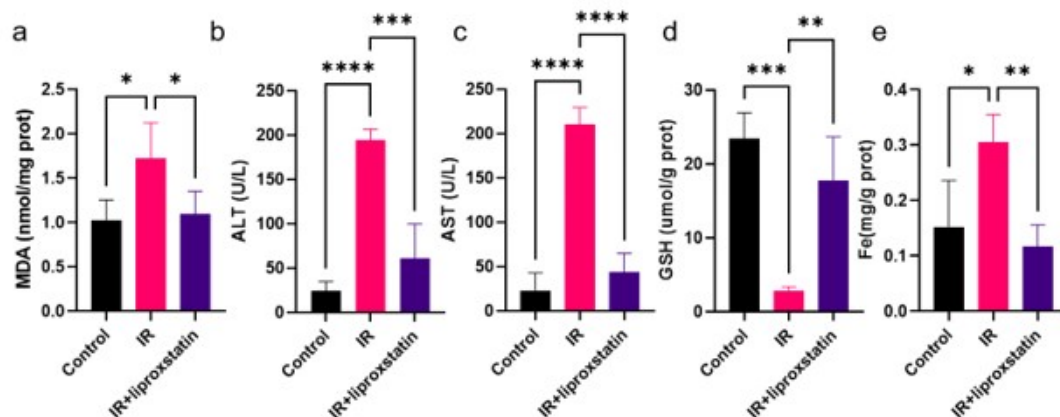
We acknowledge that imaging techniques may have limitations in providing quantitative data. However, we employed BODIPY 581/591 C11 assays as a widely accepted and commonly used method to assess lipid ROS levels. This allowed us to obtain qualitative and semi-quantitative information on the changes in lipid ROS levels in response to metformin treatment.

1. Liproxstatin may be a better drug choice or at the very least should be used to compare with the DFO data

Thank you for your suggestion. We have taken your advice into consideration and conducted an evaluation of Liproxstatin as a ferroptosis inhibitor. Our findings indicate that Liproxstatin significantly improves HIRI (Figure C). We believe that incorporating Liproxstatin in our research will provide valuable insights and allow for a comprehensive comparison with the DFO data.

Author response image 6.

Figure3 a: Liver MDA detection; b: Serum ALT level; c: Serum AST level; d: Liver GSH level; e: Liver Fe level.



1. The rationale for how GABA was selected is not clear. I am surprised that there were not more significant metabolite changes. It might be better to show a volcano plot of heatmap of the significantly changed features.

Thank you for raising an important question regarding the rationale for selecting GABA as the focus metabolite in our study. Initially, we also had concerns about the limited number of significant metabolite changes observed. However, through our comprehensive metabolomic profiling, we identified GABA as the most significantly altered metabolite following HIRI.

It is worth noting that we specifically focused on the measurement of 22 essential amino acids in our analysis. While it is possible that changes in non-essential amino acids may have occurred, we did not examine them in this study. Nevertheless, we have since used additional methods to validate the upregulation of GABA levels, and the biological effects observed support the specific role of GABA in protecting against HIRI. Based on the fact that GABA was the only significant amino acid, the volcano plot was of little significance, so we did not supplement this plot.

We appreciate your valuable input and thank you for bringing up this important issue.

1. The manuscript needs to be proofread and edited. There are a variety of typos and grammar issues throughout.

Thank you for your feedback. We acknowledge that the manuscript requires proofreading and editing, as we have identified several typos and grammar issues. We will try to ensure that the necessary revisions are made to improve the overall quality of the manuscript.

Reviewer #3 (Recommendations For The Authors):

However, I have some major concerns for the manuscript.

1. Line 26 16S rRNA and metagenomic sequencing alone can't accurately confirm the improvement effect of GABA producing bacteria on HIRI. In fact, transcriptome analysis, HPLC-MS/MS and other methods were also used in this paper, so the language expression here is not appropriate

Thank you for pointing out the language expression issue in line 26 of the manuscript. We apologize for any confusion caused. You are correct in stating that 16S rRNA and metagenomic sequencing alone may not accurately confirm the improvement effect of GABA-

producing bacteria on HIRI. In our study, we employed a combination of multiple methods, including transcriptome analysis, HPLC-MS/MS, especially detection of bacteria GABA key synthetases, PAT and GAD, to comprehensively investigate the impact of GABA-producing bacteria on HIRI.

We have revised the language in line 26 to reflect the broader range of methods used in our study to support the conclusions regarding the improvement effect of GABA-producing bacteria on HIRI.

1. The Introduction section needs to add a description of the previous research on the association between HIRI and ferroptosis

Thank you for your suggestion regarding the inclusion of a description of the association between HIRI and ferroptosis in the Introduction section. We agree that this is an important aspect to address. However, upon further consideration, we have decided to move the discussion of ferroptosis and its potential role in HIRI to the Discussion section, as it aligns better with the logical flow of the manuscript. This allows us to discuss the potential implications and future directions in a more organized and coherent manner.

1. Authors should provide quantified figure or table next to the results of western blot that are more convenient to understand.

We have revised in manuscript. (See sfigure 7)

1. In this paper, FMT experiments are used to verify that metformin remodeled gut microbiota can play a role in improving HIRI. The operation steps of FMT should be described more specifically in the method part

**What is the fecal donor information for FMT?*

**Line272 Did the IR + FMT group put the transplanted microbiota of FMT directly into the drinking water like the other treatment groups? Will such an operation affect the quality and quantification of the transplanted microbiota and lead to the loss of microbiota species? It is crucial for the authors to provide a clear and thorough clarification regarding these matters within the context of their FMT experiment.*

Thank you for your feedback regarding the need for a more detailed description of the fecal microbiota transplantation (FMT) procedure and clarification regarding the IR + FMT group in our manuscript. We appreciate your suggestions and we have taken them into consideration.

In our study, the fecal donor for FMT was obtained from mice that had been orally administered metformin. The fecal microbiota was collected and processed to remove any residual metformin before transplantation. Specifically, the microbiota for the IR + FMT group was administered through gavage, as stated in line 272. This method does not affect the quality or quantity of the transplanted microbiota, nor does it lead to a loss of microbiota species. We understand the importance of providing clear and thorough clarification regarding these matters. Therefore, we have included additional specific details of the FMT procedure in the revised version of the manuscript. We hope that this clarification addresses your concerns and provides a more comprehensive understanding of our FMT experiment.

1. *The presentation of transcriptomic analysis results in the manuscript is insufficiently comprehensive and specific, as they are solely depicted through Fig 4a. Relying solely on Fig 4a is inadequate to establish the definitive roles of the met group and FMT group in ferroptosis compared to other groups. Therefore, the authors should provide additional transcriptomic analysis results to ascertain the specific effects of the met group and FMT group in ferroptosis, as well as their comparison with other groups.*

Thank you for your feedback regarding the comprehensiveness of our transcriptomic analysis results in the manuscript. We understand your concerns and appreciate your suggestion. In our study, we have provided additional data beyond Fig 4a to support the specific effects of the met group and FMT group in ferroptosis, as well as their comparison with other groups. Specifically, in Figure 3, we have included Western blot (WB) and quantitative real-time polymerase chain reaction (qRT-PCR) data to confirm the involvement of ferroptosis in HIRI and the role of metformin in attenuating ferroptosis. Moreover, we have presented transcriptomic analysis results in Figure 3h, which includes a heatmap of genes related to lipid metabolism. These findings can strengthen our conclusions regarding the importance of ferroptosis in HIRI and the protective effects of metformin against ferroptosis. We hope that these data address your concerns and provide a more comprehensive understanding of our research findings.